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Transition between *water-in-DES* to *DES-in-water* in choline chloride–lactic acid mixtures: Implications for polyphenol extraction

Cristian Malebrán [a], Teresa Poblete [a], Daniela Millán [a,*], Rodrigo Ormazábal-Toledo [b,*]

[a] Centro Integrativo de Biología y Química Aplicada (CIBQA), Universidad Bernardo O'Higgins, General Gana 1702, Santiago, 7780272, Santiago, Chile
[b] Departamento de Química Orgánica y Fisicoquímica, Facultad de Ciencias Químicas y Farmacéuticas, Universidad de Chile, Dr. Carlos Lorca Tobar 964, Independencia, Casilla 233, Santiago, Chile

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ABSTRACT

In this article, we present an experimental and theoretical study on the structural and dynamic solvation of aqueous mixtures of Choline Chloride:Lactic Acid (in a 1:2 ratio) deep eutectic solvents and their impact on polyphenol extraction from Luma Apiculata. We use a combined experimental and molecular dynamics approach to analyze key solvent properties through FT-IR, thermogravimetric analysis (TGA), Kamlet-Taft solvatochromic parameters, and molecular dynamics. Our results reveal that up to 67 mol% water, DES maintains its nanostructure, optimizing polyphenol extraction; however, beyond this threshold, water disrupts the DES framework, leading to a more aqueous-like solvation environment. Polyphenol extraction remains efficient at higher water content, suggesting DES components reorganize affecting solvation dynamics. These findings provide critical insights into the hydration limits of DES, emphasizing the need to consider water as an active participant in solvent design for bioactive compound extraction.

1. Introduction

Deep eutectic solvents (DES) have been extensively used for the extraction of various bioactive compounds, including polyphenols, [1,2], showing better efficiency compared to traditional solvents, such as water, ethanol or methanol [3–5]. DES are recognized as a class of sustainable solvents due to their composition, which often consists of natural, non-toxic compounds. Additionally, they exhibit properties such as low vapor pressure, low synthesis cost, etc., enhancing their green credentials [6]. Generally, DES are prepared by mixing a hydrogen-bond acceptor (HBA) with a hydrogen-bond donor (HBD) in specific molar ratios to form a eutectic mixture at room temperature. [6] Their physicochemical properties can be tailored by varying the nature and ratio of their hydrogen-bonding components, making them versatile for several applications, such as synthesis of functional materials [7], electrochemistry [8], separation process [9], and pharmaceutical applications [10–12]. To study the structural changes when varying the HBA and HBD composition, several approaches have been used, such as Fourier-transform infrared spectroscopy (FT-IR) [13], conductivity [14], thermogravimetric analyses (TGA) [15], and differential scanning calorimetry (DSC) techniques [13]. On the other hand, their specific interactions can be characterized through theoretical studies [16] and empirical solvent parameters, such as Kamlet-Taft solvatochromic parameters, where the use of solvatochromic probes sensitive to microscopic changes in the medium can account for these interactions [17]. Moreover, in the last decade, DES have gained attention as green and efficient solvents for extracting bioactive compounds from plants, such as polyphenols [2], flavonoids [18], and terpenoids [19] due to their biodegradability, low toxicity and tunable properties. In fact, DESs enable selective extraction by adjusting their hydrogen-bonding components.

However, understanding the solute-solvent interactions and solvent rearrangements in the extraction process from plants is a challenging task because of the influence of multiple factors, including polarity, solid-to-solvent ratio, pH, water content, and viscosity [20]. In particular, the high viscosity of DES is another obstacle, as it hinders mass transfer between the sample and the extraction phase. Moreover, viscosity complicates the instrumental analysis, making it difficult to characterize analytes of diverse nature [21]. To overcome this problem, the addition of water offers a quick solution by decreasing the viscosity of DES, improving mass transfer and extraction efficiency [22]. Nevertheless, while the addition of water can reduce the viscosity and enhance the extractive capabilities of DES for specific analytes, it also affects its

* Corresponding authors.
E-mail addresses: cmalebran@postgrado.ubo.cl (C. Malebrán), pteresa@postgrado.ubo.cl (T. Poblete), daniela.millan@ubo.cl (D. Millán), rodrigo.ormazabal@ciq.uchile.cl (R. Ormazábal-Toledo).

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nanostructure and the natural physicochemical properties such as polarity, density, electrical conductivity, etc., leading to the extraction of undesired molecules [23].

From a molecular point of view, water greatly affects the intrinsic structure of DES by disrupting the hydrogen bond networks between HBD and HBA components [24]. Ongoing research on this topic aims to understand not only the effect of water on the nanostructure of DES but also whether there is an upper limit of hydration beyond which these mixtures cease to be DES and become more like an aqueous DES solution [25–27]. For instance, Hammond et al. [28], analyzed the nanostructure of choline chloride/urea/water DES mixtures across a broad hydration range by experimental techniques (such as neutron total scattering and empirical potential structure refinement). They determined that water slightly disrupts the hydrogen bonding network at low hydration levels, favoring the choline-urea bonds [28]. The DES nanostructure remained up to 41% wt water, beyond which 51% wt water, water-water and DES-water interaction become dominant, turning the mixture into an aqueous solution of DES components [28]. On the other hand, Rozas et al. [29] studied the effect of water on the physicochemical properties and nanostructure of the choline chloride:ethylene glycol DES through experimental studies such as density, thermal expansion coefficient, dynamic viscosity, electrical conductivity, thermal conductivity, refraction index, normalized Reichardt's polarity parameter, and molecular dynamics (MD). They found a non-linear evolution of the properties and structure as water content increased, concluding that this solvent can be hydrated up to 30% wt, maintaining most of the features of the eutectic mixture.

Within this regard, theoretical studies can provide a deeper understanding of the role and nature of such interactions. For instance, Kumari et al. [30] obtained similar conclusions through MD to those of Hammond et al. [28]. Sapir and Harries [27] also found, through MD, a transition between a *water-in-DES* (below 30 wt% H_2O) to a *DES-in-Water* (30–50 wt% H_2O), whereas beyond 50 wt% H_2O the solution changed into an aqueous electrolyte-like mixture in the choline chloride:urea mixture [31]. Aparicio et al. recently studied the water content in choline chloride:lactic acid DES using MD simulations [16]. They found that water content in a DES-water system can reach up to 2000 ppm without affecting the structural characteristics of DES, as water occupies available spaces within the DES structure through hydrogen bonds. Beyond 2000 ppm, water molecules disruptively interfere with the DES structure.

Even though a transition between a DES and an aqueous solution has been reported before, the mechanisms governing this change are not yet fully understood. For this reason, we present a combined experimental and theoretical investigation into the choline chloride:lactic acid (1:2) based DES and their mixtures with water, focusing on their implications in the polyphenol extraction from Luma apiculata – a South American tree used in traditional medicine. Through a systematic study of DES-water mixtures, we correlated changes in extraction efficiency with variations in different physicochemical properties, including polarity, viscosity, and hydrogen-bonding interactions. We chose a well-known system we have studied (the extraction of polyphenols from Luma apiculata) to experimentally determine these interactions in the series of aqueous DES mixtures [32]. Recently, we have found promising results with different DES by changing the nature of HBD and HBA components; however, the amount of water the system could tolerate while maintaining extraction efficiency is still an unresolved question. For this reason, we prepared a set of DES-water mixtures for this work and carried out an ultrasound extraction to determine the total phenolic content (TPC) in each mixture. To give insights into the role of the solvent in the extraction process, we characterized each aqueous mixture by TGA analysis, FT-IR and Kamlet-Taft solvent parameters, complemented with MD studies to contribute to a deeper understanding of DES-water systems. In summary, in this article, we present a systematic study based on different experimental techniques, complemented with MD simulations to provide a molecular-level perspective on the structure and solvation dynamics of the DES-water mixtures. This study contributes to the ongoing discussion on the limits of hydration in DES, offering new insights into the water/DES mixtures and their impact as, for instance, extraction media.

Table 1
Composition of each DES considered in the present study and water concentrations expressed in percentage by mol and weight.

Entry	Molar ratio			%mol of water	% w/w of water
	ChCl	LA	Water		
1	1	2	0	0.0	0.0
2	1	2	1	25.0	5.3
3	1	2	2	40.0	10.1
4	1	2	3	50.0	14.5
5	1	2	4	57.1	18.4
6	1	2	5	62.5	22.0
7	1	2	6	66.7	25.3
8	1	2	7	70.0	28.3
9	1	2	10	76.9	36.0
10	1	2	15	83.3	45.8
11	1	2	20	87.0	53.0
12	1	2	35	92.1	66.4

2. Experimental and computational methods

2.1. Chemicals and reagents

All solvents and reagents used in the study were of analytical grade and used without further purification. Choline chloride (ChCl), lactic acid (LA), and MilliQ water were purchased from Merck (Chile).

2.2. DES preparation

The ChCl:LA (1:2) water-DES mixtures were prepared by magnetically stirring the three components for 3 h at 60 − 80 °C until a clear, homogeneous liquid was obtained. The amount of water in each mixture varied from 25-92% mol. The molar ratio composition of each DES and the amount of water it contains, expressed as percentage per mol of water and percentage by mass of water, are presented in Table 1. The initial water content of neat DES (ChCl:LA, 1:2) was determined using Karl Fischer titration and found to be 0.7%. A similar value was obtained from thermogravimetric analysis, as will be seen below.

2.3. FT-IR and TGA analyses

FT-IR Analyses of each aqueous mixture were performed on a Jasco 4100 instrument equipped with an ATR system. The spectra were recorded in the absorbance mode from 4000 to 400 cm^{-1} with 50 scans per spectrum with a 4 cm^{-1} resolution. Thermogravimetric analyses (TGA) were performed using a Perkin Elmer 4000 thermal analyzer with Al_2O_3 crucibles. The samples (20 mg) were heated from 25 °C to 600 °C, under nitrogen atmosphere (20 mL/min) with a heating rate of 10 °C/min. It is crucial to note that ATR-FTIR predominantly analyzes surface layers, and variations in intensity with increasing water content may partially result from dilution effects. Consequently, our analysis not only considers peak intensities but also investigates alterations in band shape and position, which indicate rearrangements in hydrogen bonding. These findings are further corroborated by molecular dynamics and solvatochromic trends (see below), providing a comprehensive understanding of the structural evolution within the DES–water system.

2.4. Solvent parameters

Stock solutions (2 mM) of 4-nitroaniline, N,N-diethyl-4-nitroaniline and Reichardt dye 500 mM were prepared by dissolving them in ethanol in precleaned amber glass vials and stored at 4 ± 1 °C. 50 μL of the probe solution was transferred from the stock solution to a 5 mL vial. The

ethanol was evaporated, and the dry residue was dissolved in the desired mixture to a final volume of 500 μL for absorbance experiments. A Jasco V-630 spectrophotometer was used to acquire the UV-Vis absorbance data. All absorbances were measured using quartz cuvettes of optical path 0.2 cm at 25 °C. All spectroscopic measurements were performed in triplicate (see Tables S5-S7). Annex S1 in the Supplementary Material describes equations used to determine the parameters.

2.5. Ultrasound-assisted extraction and determination of the total phenolic content

Extracts were prepared as follows: 100 mg of dry Luma apiculata leaves were accurately weighed and placed in a 2 mL microcentrifuge tube. Then, 1 mL of the reference (80% v/v ethanol:water) or a DES-aqueous mixture was added and stirred for 1 minute before ultrasonic irradiation (50 – 60 kHz) for 45 min at 30 – 40 °C with a Sonorex Digital 10 P Bandelin Electronic device. The resulting mixture was then centrifuged for 15 min at 12.300 G in an Eppendorf model 5804 R centrifuge, and finally, the clarified supernatant was transferred to a 2 ml tube and stored as stock solution for further analysis.

2.6. Total phenolic content (TPC) and total flavonoid content (TFC)

The TPC of the different Luma Apiculata extracts was measured by the Folin-Ciocalteu assay. Briefly, 10 μL of each diluted sample was added to the 96-well plate, followed by 50 μL of Folin-Ciocalteu reagent (10% v/v). The mixture was left to stand for 3 min in the absence of light, and then 140 μL water and 100 μL of Na_2CO_3 (20%w/v) were added. After gently mixing, the sample was left in the dark for 30 minutes at 40 °C. The absorbance of the resulting solution was read at 765 nm against the solvent blank in a JASCO UV-vis spectrometer. To estimate TPC values, standard gallic acid concentrations ranging from 1 to 100 μg/mL were used to build a calibration curve. TPC values were expressed in *mg* gallic-acid equivalents (GAE) per *g* of dry sample. All measurements were performed in triplicate.

To evaluate the TFC for the extract to Luma Apiculata, 10 μL of a $NaNO_2$ solution (5%w/v) was added to 80 μL of the sample stock solution. The mixture was incubated in the dark for 5 min. Then 10 μL of aluminum chloride ($AlCl_3$, 10%w/v) and 200 μL of NaOH 4%w/v were added to the mixture. The solution was shaken and incubated for 30 minutes in the dark at 25 °C. The absorbance of the resulting solution was read at 415 nm. Quercetin was a standard in a calibration curve with concentrations ranging from 2 to 14 g/mL. The total flavonoid content of the sample was expressed in milligrams of quercetin equivalents (mg QE) per gram of dry sample. All measurements were performed in triplicate. The values obtained for TPC and TFC are displayed in Table S1.

2.7. Molecular dynamics simulations

Classical Molecular Dynamics were performed with Gromacs 2020 [33] with the OPLS force field [34]. The atomic charges for the molecules considered in the simulations were computed with CHelp-G as implemented in Gaussian 16 at the MP2/cc-pVTZ level of theory [35]. Water was simulated with the TIP5P model. Cubic cells were generated by a random distribution of *N* molecules of the DES components with Packmol 20.3.3 [36]. All simulated systems are listed in Table S2. After being minimized with the steepest-descent algorithm, the system was thermalized with a 0.8 ns NVT simulation and a step of 2.0 fs. Constant final densities were obtained after at least 80 ns NPT simulations, which were followed by a 100 ns production run. The system temperature was kept constant at 298.15 K, with the V-rescale [37] chain thermostat with a coupling constant of 0.1 ps and the pressure was controlled at 1.0 bar through the Parrinello-Rahman barostat with a coupling constant of 2.0 ps and an isotropic box vector scaling [38,39]. All simulations were obtained with a 1.4 nm cutoff radius, using a 2.0 fs time-step with LINCS

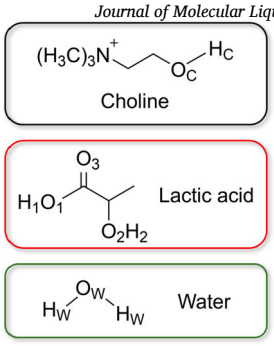


Fig. 1. Atom labeling for each studied molecule. The scheme omitted the chloride anion (Cl along the article) for simplicity.

constraint algorithm to integrate the dynamics equations. Radial and spatial distribution functions (RDF and SDF) in liquid DES as well as cluster analyses were obtained with TRAVIS [40,41] according to the labels summarized in Fig. 1. In the cluster analyses, each component of the DES was considered as a cluster. [42] Thus, a cluster is formed when, at a certain distance, a pair of components remains within a threshold distance. The distance is explicitly obtained by comparing the atom-to-atom RDF for each component.

3. Results and discussion

3.1. Extraction of polyphenols from Luma Apiculata

We have recently demonstrated that DESs effectively extract polyphenols from Luma Apiculata [33]. In our recent contribution, we screened a series of DES to identify the most suitable solvent for optimal extraction, finding that 1:2 mixture of ChCl and LA as one of the most effective DES for polyphenol extraction (with a water content of 50% w/w) due to its low viscosity and higher yields. It is well known that viscosity plays a significant role in DES performance, and the addition of water is a common strategy to reduce it [2]. Different authors have pointed out that incorporating up to 40% w/w of water in DES formulations provides an optimal balance between viscosity and extraction yield [2,43–45]. However, this raises a fundamental question of what defines the boundary between a eutectic solvent and an aqueous solution. To address this question, we propose that analyzing the extraction process in aqueous media with varying water content will offer a valuable starting point to unravel the *DES/water-in-DES/DES-in-water* transition.

To explore this relevant aspect of DES structure, in this work, we first conducted extractions of *Luma Apiculata* leaves to quantify the TPC and the TFC in aqueous mixtures of DES. The comparison of the extracting ability of phenolic compounds by pure DES and water and all the aqueous mixtures of DES are summarized in Fig. 2. A detailed description of the extraction results is also gathered in Table S1.

From Fig. 2, it is possible to distinguish similar trends in TPC and TFC plots, with distinct behavior observed in two ranges of water content: 25-77% mol and above 77% mol. Regardless of these ranges, the changes observed for TPC are more pronounced within the 25-77% mol water than for TFC in the same range of water content. For TPC, a clear maximum is observed when the extraction was performed with 63% water content, followed by a decrease in TPC content as the water concentration increases, reaching its lowest point at 77% water mol. Similarly, in the TFC plot, the maximum is observed between 57-67% water mol, but the subsequent decline is less pronounced, extending to 77% water mol.

Interestingly, above 77% mol water, the extraction of both TPC and TFC displays an intriguing response: a sudden rise reaching new maximum values for TPC and TFC content. These results demonstrate that the extraction yield can be attributed to different factors that will be discussed later. While Jiang et al. [46] suggested that an optimal water content near 60% mol water maximizes the extraction efficiency of

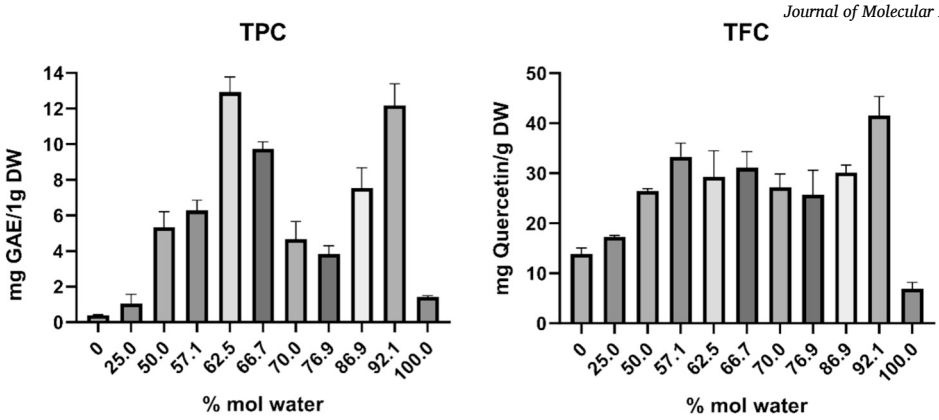


Fig. 2. Comparative total phenolic content (TPC), expressed in gallic-acid equivalents (GAE) and comparative total flavonoid content (TFC) expressed in quercetin equivalents (QE) from leaves of Luma apiculata in aqueous mixtures of DES ChCl:LA.

polyphenols, to the best of our knowledge, no articles have ventured to the origin and consequences of the water content. At this point, the question arises of whether extraction is promoted by *water-in-DES*, *DES-in-water* or both. In other words, what is happening at a molecular level within the nanostructure of DES, and how is the DES structure influencing the extraction of polyphenols? To address these questions, we studied the solvent and the rearrangement of the solvent molecules as the water content increased, merging experimental analysis with molecular simulations to gain deeper insights.

3.2. Solvation in DES

FT-IR analyses provided valuable insights into the molecular interactions and structural changes that may occur within the system upon adding water. The FT-IR spectrum obtained for the 70% mol water mixture is shown in Fig. 3(a) as a representative case. Figs. S1 to S8 display the FT-IR spectrum for all mixtures.

The FT-IR spectra reveal in the 1720 cm^{-1} region, a sharp signal was assigned to the stretching vibration of the carbonyl group in lactic acid, which decreases with increasing amount of water, accompanied by a bending-type absorption of the OH group near 1630 cm^{-1}. Characteristic absorptions of the C–N single bond in ChCl are observed at 3024 and 1482 cm^{-1}, with the latter showing decreasing intensity as the amount of water increases (see Figs. S1 to S11 in the Supplementary Material). CO stretching and CC characteristic absorptions may also be identified in the fingerprint region between $1300-1000$ cm^{-1}. There is a sharp peak in the range of $3200-3600$ cm^{-1}, where two stretching vibrations are observed, around 3200 cm^{-1} and 3400 cm^{-1}, assigned to the OH vibration of choline and the OH vibration of water, respectively [47]. This sharp peak is also observed in pure ChCl, pure DES (bold black line) and their mixtures with water (see Figs. S9 to S11). As the water content increases, the peak progressively broadens, as shown in Fig. 3(b), and the signal attributed to the OH stretching from water increasingly overlaps with the signal of the C–N single bond. This result suggests new interactions between the OH groups from water and the C–N single bond and OH group from choline cation, supporting the formation of a more complex hydrogen bond network as the water is incorporated into the system [48]. The structural characterization of aqueous DES mixtures shows characteristic bands corresponding to their functional groups. For this reason, to give more insights into the results obtained from the FT-IR analysis and its relationship with the extraction process, we obtained the atom-to-atom Radial Distribution Function (RDF) (see Fig. 4). However, considering the wide range of possible associations in the studied DES, our analysis will focus only on those most likely to be influenced by water addition based on the FT-IR results. The leading associations considered are highlighted in green in Fig. 4.

Interestingly, the probability of finding LA or water molecules surrounding the Ch cation follows a similar trend across different water

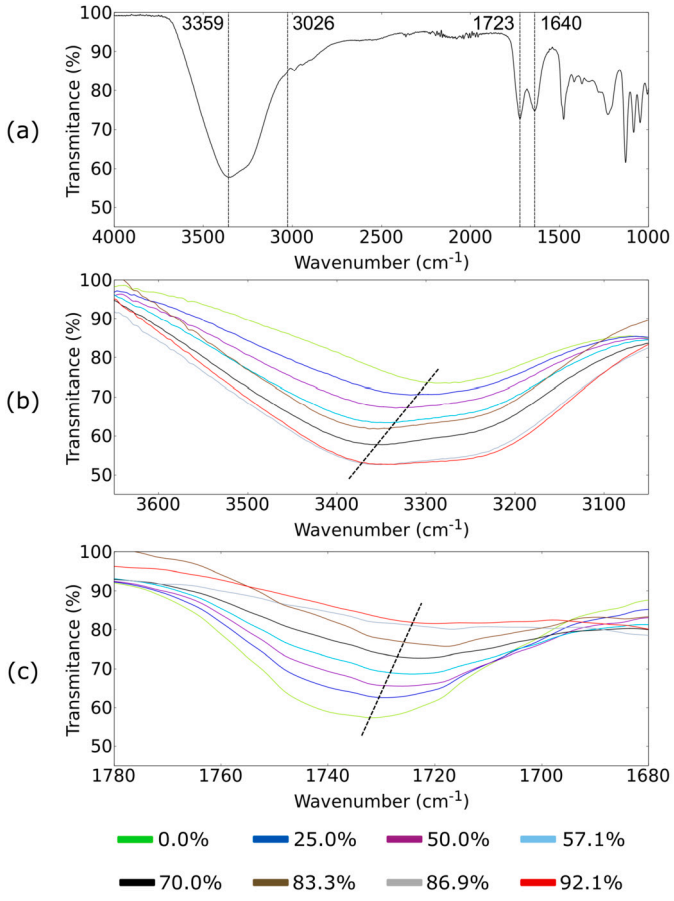


Fig. 3. FT-IR spectra of (a) DES with 70% water mol, (b) aqueous DES mixtures within the range 3700 to 3000 cm^{-1}, (c) aqueous DES mixtures within the range 1820 to 1680 cm^{-1}.

proportions (see Fig. 4(a) and (b)). In the pure DES, the absence of water naturally promotes stronger associations between the H_C atom in choline and O_3 in lactic acid (see Fig. 4(a)). However, the atom-to-atom RDFs obtained between H_C in choline and O_1 or O_2 in LA, present lower $g(r)$ values at similar distances (see Figs. S12 and S13). Fig. 4(b) and (c) reveal an unexpected trend: contrary to initial expectations, the $g(r)$ values for water surrounding H_C in Ch (Fig. 4(b)) and O_3 in LA (Fig. 4(c)) decrease as the water content increases. This behavior can be attributed to the solvation process in the mixtures, where increasing water content enhances the system entropy due to the solvation of chloride anions [49,50]. Supporting evidence for this phenomenon is presented

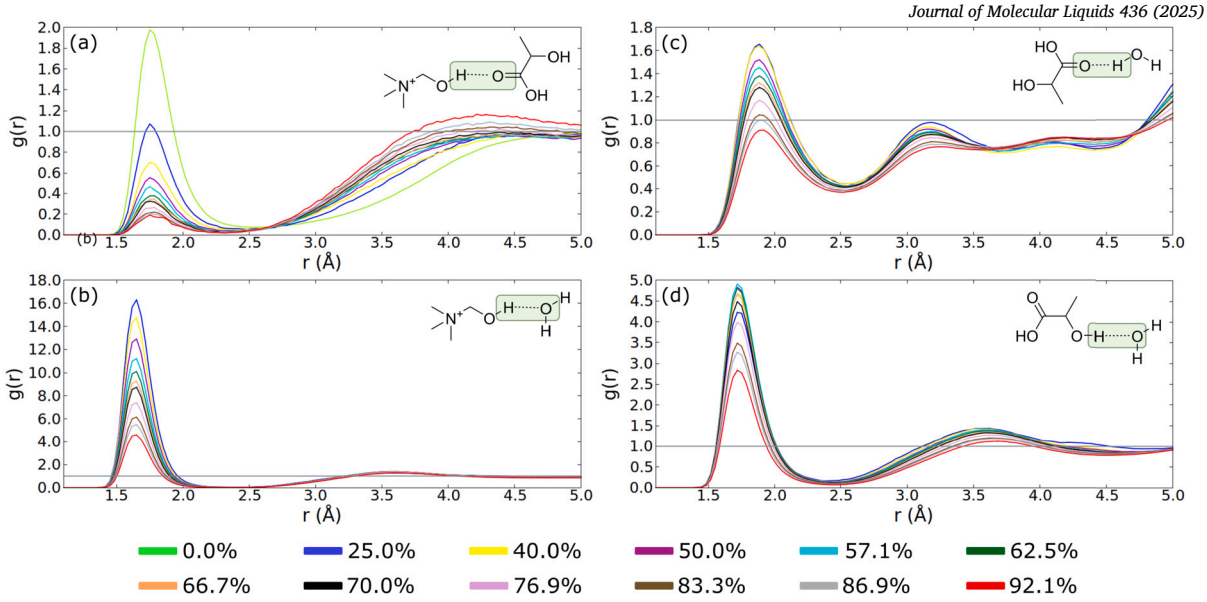


Fig. 4. Atom-to-atom Radial Distribution Functions (RDF) were obtained for the different DES-water mixtures from MD production runs. Each plot displays the most relevant associations between atoms from the IR spectra: (a) H_C in Ch and O_3 in LA, (b) H_C in Ch and O_W in water, (c) O_3 in LA and H_W in water, and (d) H_2 in LA and O_W in water. The color code presented is related to the water content in each solvent.

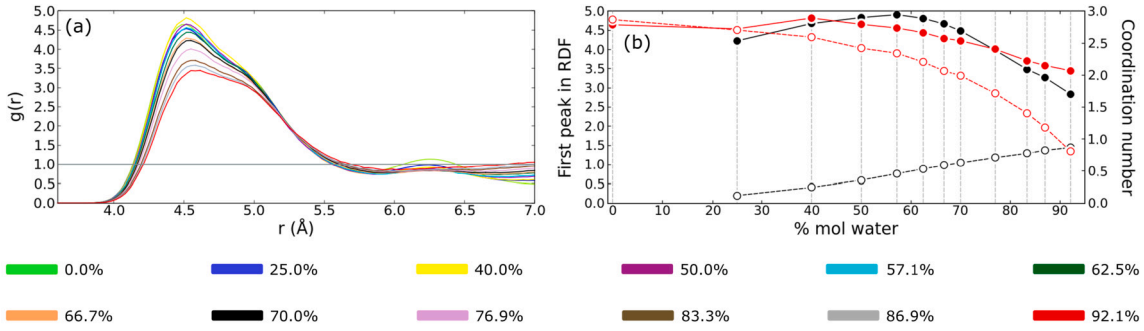


Fig. 5. (a) Atom-to-atom Radial Distribution Function (RDF), obtained for the different DES-water mixtures, between N in choline and chloride. (b) Variation of g(r) value (solid line) and coordination number (dashed line) with the water content from the RDFs presented in (a) (in red) and Fig. 4(d) (in black). The coordination number was obtained by integrating $g(r)$ up to the first minimum.

in Fig. 4(d), which shows a gradual increase in the $g(r)$ value for interactions between H_C in LA and hydrogen atoms in water (O_W) from 25.0 to 57.1% mol of water, followed by a decline in the $g(r)$ value from 62.5 to 92.1% mol of water.

The role of Cl^- on the structure of DES has been previously analyzed, emphasizing its dual role, that is, as an agent for (i) promoting mixing entropy [50] and (ii) maintaining an equilibrium of the water content within the DES [32] structure. Indeed, in the pure DES, chloride is most likely to be found in the neighborhood of N^+ of Ch, due to electrostatic interactions (see Fig. 5(a)). However, as water content increases, a $[Cl_2(H_2O)]^{2-}$ cluster forms, leading to a structural reorganization in which water molecules are located between Cl^- and Ch and/or LA (see Fig. S27).

Moreover, the RDFs in Fig. 5(a) show that the first maximum of $g(r)$ decreases with the increment of water content in the solvent. Additionally, Fig. 5(b) reveals that RDF peaks of Fig. 5(a) remain nearly constant up to 50% of water content; after that, a slight decline is observed (solid red line). However, the coordination number (i.e., the average number of chloride ions surrounding N^+ in choline) decreases from 4.6 in pure DES to 0.8 ions on average in 92.1% mol of water. The data support the formation of the $[Cl_2(H_2O)]^{2-}$ cluster, increasing the system's entropy. The observed increase in entropy may be attributed to changes in the local environment of the chloride anion as the system transitions from pure DES to the aqueous mixtures. In the pure DES, the chloride anion is tightly integrated into the hydrogen bond network formed by the DES constituents, choline and lactic acid. The high viscosity of the pure DES restricts the molecular mobility of the chloride anion, limiting its degrees of freedom. However, when water is introduced into the system, a competitive interaction arises between the water molecules and the DES components for association with the chloride anion. This competition leads to a gradual loosening of the rigid DES structure. As a result, the mobility of chloride is increased due to enhanced solvation by water molecules. This increased mobility is related, within this context, to a system's overall entropy, supporting the previously proposed hypothesis that the chloride anion plays a crucial role in the structural transition of the system to an aqueous solution. This interpretation is consistent with our molecular dynamics simulations, which revealed changes in the radial distribution functions and coordination numbers of chloride as water content increases, further suggesting the key role of chloride in the structural dynamics of DES. [50]

As a reference, Fig. 5(b) also displays the value of $g(r)$ at the first peak in the RDFs presented in Fig. 4(d) (solid black line), highlighting changes in the aqueous environment surrounding H_2 in LA, reaching a maximum near 60% mol of water. Additionally, the smooth in-

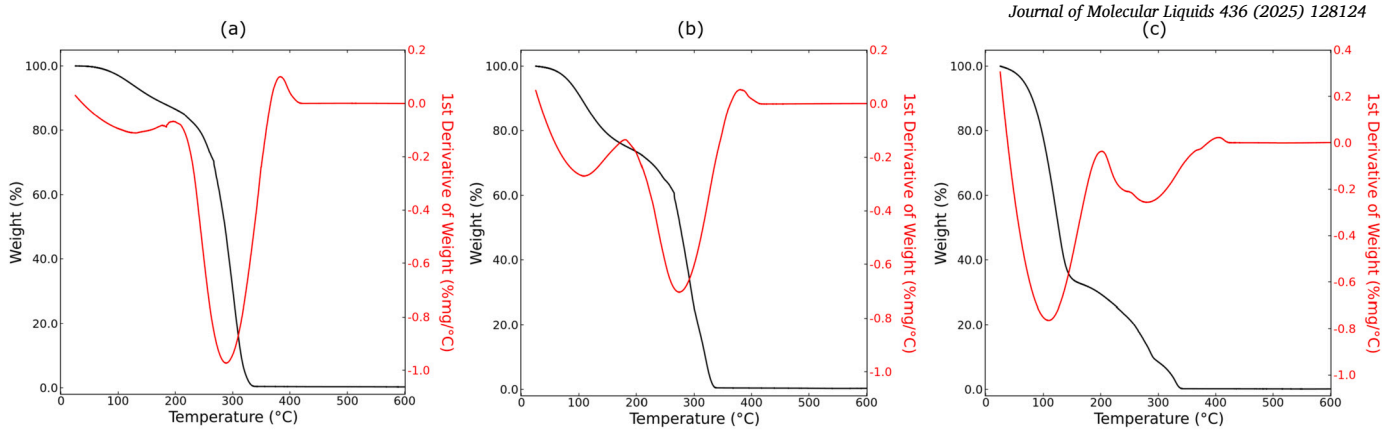


Fig. 6. Dynamic TGA curves obtained with a 10 °C/min heating rate (black). The first derivative of the heating curve is presented in red. (a) 25% of water, (b) 66.7% of water, (c) 92.1% of water.

crease in the coordination number may be indicative of a transition between *water-in-DES* to *DES-in-water*. Even though the trend observed in Fig. 5(b) resembles the results obtained in the extraction plots in Fig. 2, the effects of increased water content are far from being clearly understood. Therefore, the next task is to analyze the role of the non-interacting water content on the system.

3.3. Unbound water modulates DES structure

Thermogravimetric analysis (TGA) is a useful tool for analyzing the extent of hydrogen bond weakening and bond dissociation between HBA and HBD in DES [51]. In fact, Florindo and coworkers have related the onset temperature (T_{onset}) and peak temperature (T_{peak}) as characteristic parameters of DES stability [52]. Specifically, T_{onset} represents the highest temperature at which a DES remains in its liquid state without decomposition, while T_{peak} corresponds to the temperature at which the maximum decomposition occurs [48]. The thermal behavior of the pure DES and its water mixtures is illustrated in Figs. S14 to S25, where two zones may be identified: one from pure DES up to 50% water mol characterized by a single decomposition step and another from 57% water mol onward, where two-step decomposition process is observed. For easier comparison, the upper panel of Fig. 6 compares pure DES, 70.0% and 92.1% water mol as representative examples. Additionally, the T_{onset} and T_{peak} values for the aqueous mixtures of DES are displayed in Table S4.

We selected a subset of three DES as representative cases of the three regimes mentioned above (see the upper panel of Fig. 6). At lower water content, the thermograms are quite similar, showing comparable behavior, but both T_{onset} and T_{peak} shift to lower temperatures as the water content increases (Table S4 shows the T_{onset} and T_{peak} values for each mixture). On the other hand, upon reaching 66% water mol, a two-step TGA is more pronounced and a new shift in the T_{peak} values is also observed. These results suggest changes in the way water interacts within the DES at this composition. As Hammond et al. stated, this response may be linked to the *water-in-DES* to *DES-in-water* transition [28]. Interestingly, from 70% water mol, a small shoulder close to 250 °C appears, which is more clearly visible by analyzing the first derivative of the plot (red lines in the upper panel of Fig. 6) and becomes more pronounced at 87% and 92% water mol. This shoulder suggests a possible saturation point of the hydration of DES, which reaches its maximum at 67% water mol. Below 67% water mol, DES interacts with water to form a three-dimensional hydrogen bond network, incorporating water into the nanostructure of DES, leading to less unbound water in the medium. Between 67% to 70% mol of water, the loss of water increases around 4% in mass, suggesting that the DES mixtures become saturated, resulting in a sharp increase in water loss compared to other mixtures. The lower panel in Fig. 6 clearly shows this phenomenon with a snapshot from the MD simulations. Notably, water is homogeneously distributed in the simulation box in DES with low water content and the mixture with 66% mol water. However, in 92.1% mol water, excess water disrupts the hydrogen bond network and leads to clusters of Ch and LA surrounded by water. This structural reorganization may explain the anomalous results observed in Fig. 4(b) and (c). A similar pattern is found in a cluster analysis performed with TRAVIS from the MD simulation data. Fig. 7 presents the cluster count function (CCF) considering the formation of clusters between all components in the mixtures. The CCF shows that, at short distances, the number of clusters corresponds to the total number of particles considered in the system whose values, for simplicity, were normalized to 100. As the cutoff distance increases, the cluster size grows while the number of observed clusters of a given size decreases. Finally, at sufficiently long distances, the CCF approaches unity, indicating that the whole system is considered as a single cluster. The evolution of the number of clusters indicates the homogeneity of the system [42,53].

Interestingly, Fig. 7 displays similar responses to TGA plots (see Figs. S14 to S25), displaying a three-zone trend. The first zone corresponds to the pure DES or a eutectic mixture (green line), showing a decreasing melting point with respect to the pure components. At lower water concentrations, say up to 50% of water, the *water-in-DES* mixture is formed. In the range near 75% of water, the mixture may be classified as an "optimal" DES, while at higher water content, a *DES-in-water* adduct is observed. Notably, even though this classification was made from molecular dynamics results, it may be easily related to the extraction performance observed in Fig. 2. Indeed, the eutectic mixture presents poor performance (due to its high viscosity) that smoothly increases in the *water-in-DES* zone. The best extraction yields were obtained in the mixtures transiting to the *DES-in-water* zone, with, contrary to the expected, outstanding performance in the mixture presenting the highest excess of water.

An intriguing result appears related to water loss in the TGA analysis. Fig. 8 shows the atom-to-atom RDF obtained for O_W (oxygen in water) and H_W (both hydrogen atoms in water), i.e. water surrounding water molecules. The first peak intensifies with increasing water content up to 70% water mol but dramatically decreases at 92.1% water mol. However, the coordination number consistently increases across all mixtures. These findings may be related, on the one hand, to different organizations within the solution and, on the other hand, to non-interacting water content. Indeed, as shown in Fig. 8(b), a linear correlation was obtained by relating the mass of water lost from the bulk in each DES (see Table S3 and the coordination number of the RDFs in Fig. 8(a). While initially surprising, this result highlights an essential consideration: the addition of water must be considered not just as an innocent

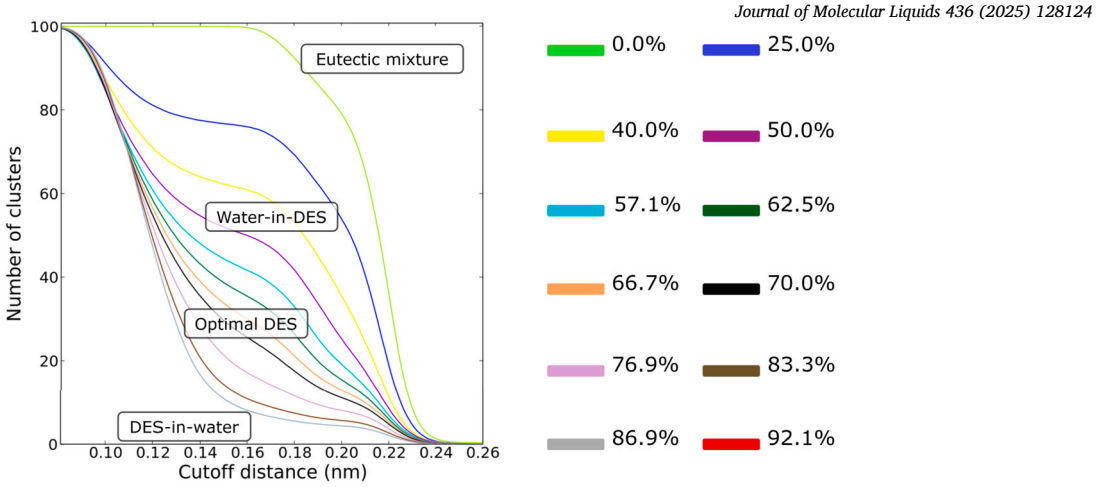


Fig. 7. Cluster count function of the DES components for each water mixture. The cluster count was normalized to 100 in all cases. The color code is the same as in Fig. 4.

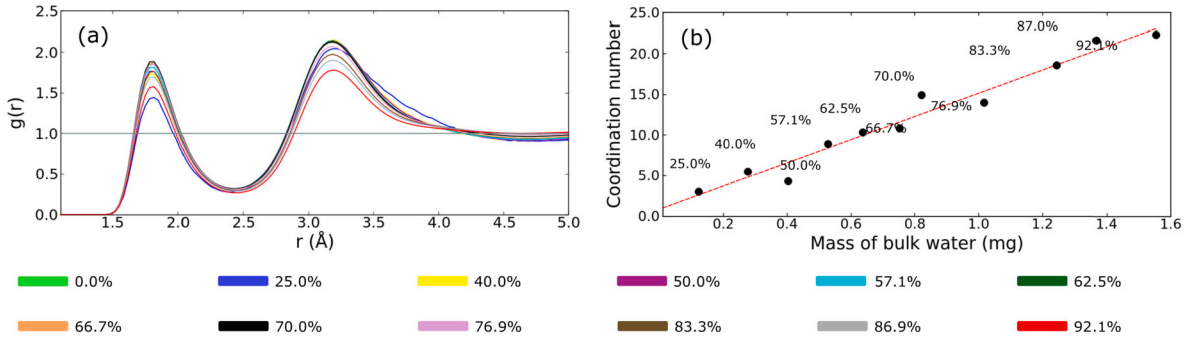


Fig. 8. (a) Atom-to-atom RDF obtained for O_W in water and H_W atoms in other waters. (b) Linear relationship obtained for the variation of mass of bulk water and the coordination number (up to the first minimum) in (a).

way to control the viscosity but as an additional component of the media that will control the performance of the solvent in any process. In other words, adding water determines the nanostructure of the solvent and, consequently, its macroscopic responses and performance.

A final remark on the extraction results for 87.0 and 92.1% mol water mixtures deserves mention. As stated above, these mixtures contain the highest unbound water levels, suggesting a lower extraction yield. However, the low solubility of polyphenols and flavonoids in water could be countered by a higher association with the Ch-LA pair in the solution. At such high water content, the original DES components may form new DES-like structures with the extracted polyphenols and flavonoids, creating favorable local environments for extraction. Although speculative, this idea is supported by recent studies on the synthesis of natural DES based on flavonoids. [54], which may be naturally obtained during the extraction process at higher water content. In other words, in the higher water content mixtures, the presence of flavonoids or polyphenols may synergistically assist the formation of a new DES based on its association with choline and lactic acid. These specific considerations deserve to be considered in upcoming research because, in our opinion, they are beyond the scope of this article.

In this sense, our findings point out that the critical hydration threshold of 67% mol water in the ChCl:LA system is in line with previous studies. However, our study provides a comprehensive molecular view of the *water-in-DES* to *DES-in-water* transition and its influence at a macroscopic level. In fact, this contribution significantly broadens the range of dilution, revealing how the ChCl:LA DES behaves at much higher water contents and identifying the critical transitions that occur as water content increases.

3.4. The transition from water-in-DES to DES-in-water

A final word related to the interactions between the components of the DES and water deserves mentioning. The Kamlet-Taft parameters (α, β, π^*) serve as molecular probes to characterize the solvation properties of a given solvent and were obtained for the water-DES mixtures [55]. In Kamlet-Taft's description, α represents the solvent's hydrogen bond donating ability, β the hydrogen bond accepting ability, and π^* polarity/polarizability. The Kamlet-Taft parameters can be determined for any solvent or solvent mixture. For mixtures, these parameters reflect the combined effect of different factors and components, allowing a comparison among different mixtures. Aqueous mixtures of variable composition can reveal specific trends, such as medium properties changes with the addition of a particular component (water, for instance) [56]. The solvatochromic values obtained for the aqueous DES mixtures are displayed as the correlations with increasing water content in Fig. 9 (raw data is summarized in Tables S5-S7 in the Supplementary Material).

The response presented in Fig. 9 is in line with the previously discussed effects of water on the nanostructure of DES. The trends are quite clear; all three solvatochromic parameters follow a smooth trend from pure DES up to 60-70 mol% water, consistent with the integration of water molecules into the DES network. In this range, α gradually decreases, indicating a progressive weakening of the DES hydrogen-bond network due to water incorporation. Simultaneously, β and π^* exhibit an initial increase, suggesting enhanced solvation dynamics driven by water-mediated interactions. However, beyond this threshold, the system appears to be perturbed. These patterns closely align with those

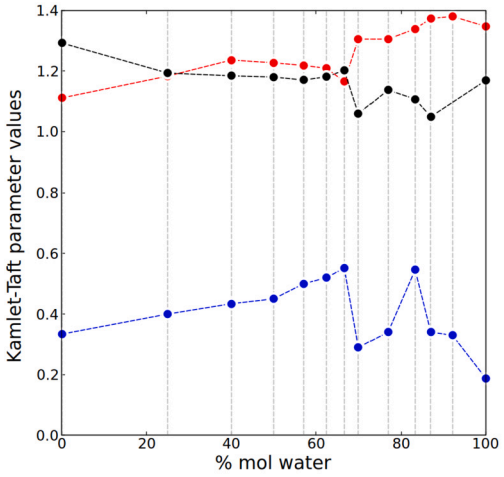


Fig. 9. (a) The solvatochromic Kamlet-Taft parameters α (black), β (blue), and π^* (red) were obtained for each DES considered in the present study.

reported elsewhere for ChCl:glycerol and ChCl:lactic acid (both 1:2) across the full range of water content. [57,58] This similarity reinforces the discussion presented above that water progressively disrupts the DES hydrogen-bond network (decreasing α), while the basicity of choline chloride (β) remains significant. At the same time, the rising π^* indicates a shift toward a more aqueous-like, polar environment. This consistency across systems strengthens and supports our conclusion of structural reorganization driven by hydration.

While Kamlet–Taft parameters provide useful semi-empirical insight into solvent polarity and hydrogen bonding, it is important to note their limitations in structured or heterogeneous systems like DES–water mixtures. The measured values reflect the local environment of the solvatochromic dye, which may be partitioned into specific domains and not fully represent the bulk properties of the solvent. As such, these parameters should be interpreted as qualitative indicators of solvation trends rather than absolute descriptors of the entire system. For this reason, the solvatochromic data were complemented by molecular dynamics simulations, which offer a more physically grounded view of intermolecular interactions and network reorganization. Indeed, as observed in both experimental and theoretical studies, water initially integrates into the DES matrix, occupying available hydrogen-bonding sites and gradually modifying its physicochemical properties. However, beyond a critical hydration level—consistent with the 67 mol% water thresholds identified here—disruptions in the hydrogen-bond network become more pronounced, leading to structural rearrangements such as cluster formation, as seen in MD simulations. This transition reinforces the idea that, at high hydration levels, DES properties evolve into an aqueous-like system, supporting the concept of a *DES-in-water* regime, suggesting a fundamental shift in solvation dynamics at high water content. These findings demonstrate that water plays an active role in modulating the solvation environment, influencing both extraction efficiency and solvent behavior at the molecular level.

4. Concluding remarks

This study provides a comprehensive understanding at the molecular level of the transition between *water-in-DES* to *DES-in-water* and its impact on the extraction of polyphenol. Through a combined experimental and theoretical approach, we demonstrate that water plays a dual role: initially integrating into the DES structure, stabilizing the hydrogen-bond network, and optimizing extraction properties, but eventually disrupting this organization at higher concentrations. Our findings show that DES maintains its structure up to 67 mol% water, with water modulating viscosity and solvation. However, beyond this threshold, a shift occurs characterized by the formation of choline-lactic acid clusters surrounded by non-interacting water, as evidenced by TGA, RDF, and KT parameters.

This structural evolution directly influences the extraction efficiency of polyphenols. While a well-balanced DES-water mixture enhances the solubility and mass transfer, excessive water disrupts the DES framework, altering its physicochemical properties and leading to an aqueous-like system. Polyphenol extraction becomes efficient again in highly hydrated systems, suggesting that DES composition changes, contributing to solvation dynamics despite the predominant aqueous environment. This finding reveals the complex interplay between solvent composition and extraction performance, highlighting the need for further investigation into the specific mechanisms and structures at the molecular level in diluted DES systems.

This multi-faceted approach allows us to correlate molecular-level changes with macroscopic properties like extraction efficiency, providing a more holistic understanding of the system's behavior across a wide range of water contents. These findings provide valuable insights into designing and applying DES-based solvents for extracting bioactive compounds. In summary, instead of considering water merely as an innocent diluent, it should be recognized as an active component that influences the solvation environment, the molecular structure of the solvent and as an agent that could hinder the solvent properties (as an extracting media, for instance). The insights presented in this contribution refine the understanding of the water-induced structural transition in DESs and provide a more comprehensive view of the water–DES network, which is especially relevant for applications requiring highly diluted systems. Future research should emphasize the role of water content in other DES formulations and model compounds, further improving our understanding of hydration limits in deep eutectic solvent applications.

CRediT authorship contribution statement

Cristian Malebrán: Formal analysis, Data curation. **Teresa Poblete:** Validation, Formal analysis, Data curation. **Daniela Millán:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Conceptualization. **Rodrigo Ormazábal-Toledo:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.molliq.2025.128124.

Data availability

Data will be made available on request.

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